MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Advanced Level
Higher 2

ECONOMICS

9757/01

Paper 1

October/November 2022

2 hours 15 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question.

Extract 3: Australian economy will need up to A$90bn extra after coronavirus shock

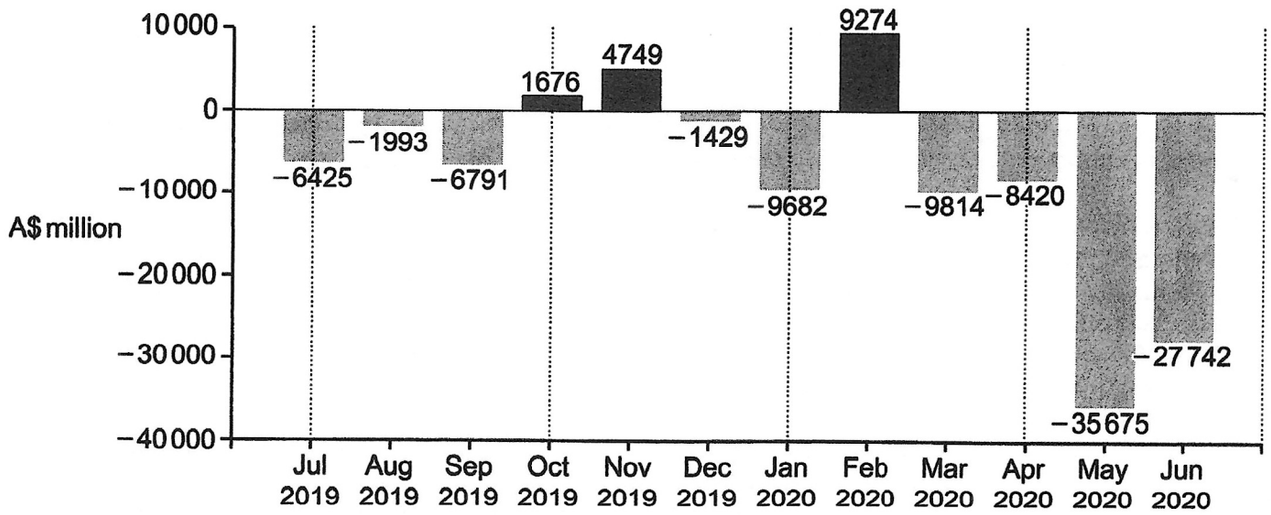
The Australian government needs to spend an extra A$70 billion to A$90 billion on stimulus and support measures to help the nation recover from the biggest economic shock since the Second World War, according to a new independent report by the Grattan Institute. This sets out a wide-ranging agenda for the next six months. It argues that the government needs to phase out virus-related emergency support measures more gradually than currently planned, to avoid slowing a recovery. It also calls on the government to introduce other forms of economic stimulus including rent assistance and childcare subsidies. With Australia facing the effects of a deep global recession, the report notes that government spending to reduce the impact has already exceeded more than A$160 billion to support households and businesses. Many of these support measures are due to end in September 2020.

The government has indicated further targeted assistance to sectors that require longer-term support, including aviation and tourism. The report warns that further stimulus is needed if the government wants to get unemployment back down to 5% or below by mid-2022, for example by spending on services, infrastructure and social housing construction.

The report says now is not the time to 'panic' about increasing government debt levels. 'Assuming interest rates remain low, the Australian government should be in no rush to balance its budget while the economy remains weak. Failing to provide support will mean many Australians remain unemployed for longer'. It also says the Reserve Bank should be prepared to use monetary policy tools more aggressively.

Source: Daniel Hurst, The Guardian, *28 June 2020*

Figure 1: Australian Budget Balance*, July 2019 to June 2020



*The budget balance is the difference between government spending and revenue.

Source: Tradingeconomics.com I Reserve Bank of Australia

(a) Use a supply and demand diagram in each case to explain why:

(i) the price of fresh fruit and vegetables in Australia has risen as a result of the drought. [2]

(ii) international tourist arrivals into Australia are expected to fall as a result of the bush fires. [2]

(b) With reference to the data, explain **one** possible reason for the change in Australia's budget balance from February to June 2020. [2]

(c) With reference to Extract 2, explain why a nominal rate of interest of 0.25% in March 2020 would be described as being 'negative' in real terms. [2]

(d) Explain how a negative real interest rate is likely to affect savings by consumers and the exchange rate in Australia. [4]

(e) Given the weakening of Singapore's exchange rate, discuss whether a stronger exchange rate would be of overall benefit to Singapore when 'the global economy is in deep recession' (Extract 4). [8]

(f) Discuss whether fiscal policy is the most effective way to bring unemployment down in Australia. [10]

[Total: 30]

Strict European Union (EU) carbon-emissions rules are forcing car makers to accelerate their plans to switch to producing electric vehicles, or face billions of euros in fines. The EU's targets for 2030 will mean that between 7 million and 10.5 million battery-powered cars will be on Germany's roads by the end of the next decade. Volkswagen's first mass-market battery-powered family car is rolling off production lines in Germany.

The cost of job losses in Germany, estimated at €100 000 per job, forces companies to consider reskilling programmes or wait for employees to retire. Yet even the German car industry predicts that the shift to electric cars will cause the loss of 70 000 jobs in the near future, as their assembly is far less labour-intensive.

Industry insiders claim that there is still reason to be optimistic. The engineering expertise that has helped Germany's car makers survive for 130 years is expected to carry them through the wave of disruption that looms on the horizon. The German car industry is expected to become a pioneer in electric car production.

Source: Financial Times, *30 November 2019*

Extract 7: Bloomberg Innovation Index: Germany replaces South Korea as most innovative nation

Table 1: 2020 Bloomberg Innovation Index (top 12 of 50 economies shown)

Economy	2020 rank	2019 rank	R&D intensity	Productivity	Tertiary education efficiency
Germany	1	2	8	18	26
S. Korea	2	1	2	29	16
Singapore	3	6	12	4	1
Switzerland	4	4	3	14	17
Sweden	5	7	4	19	13
Israel	6	5	1	15	32
Finland	7	3	10	9	24
Denmark	8	11	7	6	31
US	9	8	9	12	47
France	10	10	13	16	20
Austria	11	12	6	13	12
Japan	12	9	5	35	30

In economic terms, innovation describes the development and application of ideas and technologies that improve goods and services or make their production more efficient. Innovation is an essential driver of economic progress that benefits consumers, producers and the economy as a whole. It has two main aspects. Product innovation involves the introduction or improvement of goods or services. Process innovation involves the introduction of improved production techniques.

The Bloomberg Innovation Index uses seven criteria to rank 50 economies. These criteria include research and development (R&D) spending, productivity and tertiary efficiency.

(a) Distinguish between a rise in total output and a rise in productivity. [2]

(b) Explain **one** reason why a firm with significant market power and long-run excess profits might choose to spend large sums on research and development (R&D). [2]

(c) Explain, with reference to Extract 8, what might be the opportunity cost of the S$19 billion invested by the Singapore government to build the country into a global R&D hub. [2]

(d) Explain how the creation of an innovation culture in Singapore is likely to have benefited the Singapore economy. [6]

(e) Discuss whether subsidising the purchase of electric cars would improve the efficiency of resource allocation in the market for transport in Singapore. [8]

(f) Discuss the extent to which government policy can influence a country's comparative advantage in a good or service. [10]

[Total: 30]

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9757/02

Paper 2 Essays

October/November 2022

2 hours 15 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **three** questions in total, of which **one** must be from Section A, **one** from Section B and **one** from **either** Section A or Section B.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section B

One or two of your three chosen questions must be from this section.

4 During the pandemic, there was a fall in investment and a rise in unemployment. Innovation-based investment is a good way to stimulate the economy but, unless appropriate government policy action is taken, this might lead to another type of unemployment.

(a) Explain how a fall in investment might lead to one type of unemployment while a rise in innovation-based investment might lead to a different type of unemployment. [10]

(b) Discuss the extent to which a rise in Singapore government spending on investment in major capital projects and in human capital are **both** required to reduce unemployment in a post-pandemic world. [15]

5 A low rate of inflation is a key macroeconomic policy objective for most governments. During the first quarter of 2020, interest rates in most countries throughout the world fell to very low levels.

(a) Explain why a rise in interest rates is used as a macroeconomic policy tool to control inflation in some countries but not in Singapore. [10]

(b) Discuss whether a change in interest rates in other countries is likely to have a significant impact on Singapore's domestic and external economy. [15]

6 Singapore's Prime Minister has stated that 'Globalisation will be under pressure, but it is imperative for countries to cooperate, for businesses to operate across many geographies, to tap resources, to bring skills and talents and experiences together, and then serve markets all around the world.'

Source: pmo.gov.sg/Newsroom/PM-interview-with-BBC-for-Talking-Business-Asia, 2 March 2021

(a) Explain the benefits and costs of globalisation to Singapore. [10]

(b) Discuss the most appropriate policy measures that the Singapore government should take to increase the benefits and reduce the costs of globalisation. [15]

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Answer **three** questions in total.

Section A

One or two of your three chosen questions must be from this section.

1 Economists usually begin their analysis of decision-making by firms by assuming that the objective of a firm is to maximise its profit. In reality, however, there are many different objectives that a firm might adopt.

(a) Explain the likely effects on a firm's price and output when its objective changes from profit maximisation to profit satisficing. [10]

(b) Discuss the most appropriate strategy that a firm could adopt if its objective was to reduce the competition that it faces. [15]

2 The Singapore government announced in the 2018 Budget that the goods and services tax (GST) would rise from 7% to 9% sometime between 2021 and 2025. The intended consequence of this change is to raise tax revenue. However, following the coronavirus (Covid-19) outbreak in 2020, the incomes of many households fell.

(a) With the aid of diagrams, explain why an increase in GST and a fall in the incomes of many households are **each** expected to cause a fall in expenditure on luxury goods. [10]

(b) Discuss whether this increase in GST is likely to raise tax revenue and whether it will lead to unintended consequences. [15]

3 Street lighting is considered to be a public good. However, there are also negative externalities resulting from the generation of electricity for the lighting on the environment and the effect of bright street lights on wildlife.

(a) Explain **two** different reasons for the market failure associated with the provision of street lighting. [10]

(b) Discuss the extent to which a government should intervene in the market to ensure that the benefits of street lighting can be obtained while minimising the negative impacts. [15]

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In 2020 Germany was ranked in first place after a period of six years when South Korea was ranked first. The German manufacturing sector is still highly competitive and a source of innovation. However, there are several reasons why Germany should not be complacent about its overall innovation ranking. Innovation in the German services sector is much less impressive and about a third of R&D spending is in the car industry. This means that if the car industry is disrupted it could decrease Germany's overall strength and innovation performance. Lack of innovation in the services sector in Germany is an increasing worry, especially as the global economy shifts more towards services and away from manufacturing. The German government would be well advised to use its on-going fiscal surplus to invest and safeguard Germany's role as an innovator in a wider range of economic sectors.

Singapore has risen in the index to third place overall. It was in sixth place last year, but has experienced productivity and manufacturing gains, while it retains a best-in-world ranking for tertiary-education efficiency. Innovation is a critical driver of growth and prosperity. China has moved up the rankings, but the US's fall in the index illustrates that trade tariffs are not going to maintain its economic advantage if there is a lack of investment in education and research that can lead to innovation.

Source: Michelle Jamrisko, Wei Lu, Bloomberg, 18 January 2020

Extract 8: Creating an innovation culture – Singapore's formula to becoming a regional tech hub

For some years the Singapore government has put in place policies designed to make itself into an innovation and start-up hub in the region. Singapore has also become a magnet for foreign technology firms, hosting regional headquarters and engineering hubs for many of the world's leading companies. A major British electronics company said earlier this year that it would be moving part of its operations to Singapore. In addition, several Chinese companies also have R&D centres in the country.

For a small country with few natural resources and a population of just 5.6 million, how did Singapore transform itself into a leading centre for technology and innovation? It comes down to creating what a Singapore minister called 'enabling conditions', such as policies that are friendly to businesses, a coordinated approach between universities and private companies to provide a trained work-force, and a high quality of life to attract globally-mobile entrepreneurs and top-grade talent who can call anywhere home. The country is also helped by the continuity of government policies and a highly educated local workforce. Singapore's status as a financial centre has also made it easy to attract those with capital looking to invest in Southeast Asia. Investments in advanced technology sectors have made it a base for high-end manufacturing, capable of developing electric cars, for instance.

The Singapore government has invested S$19 billion to build the country into a global R&D hub. In March, an additional S$500 million was set aside to expand investments in a range of new technologies. The Singapore government also takes the protection of intellectual property seriously, making it an attractive place for companies to undertake R&D.

Although it has an abundance of capital, Singapore has faced talent shortages, especially with a small population compared with major economies like the US and China. Attracting foreign talent is one solution, but Singapore has also put in place schemes to train locals in the skills needed for its technological goals.

Source: Zen Soo and Chua Kong Ho, South China Morning Post, *7 September 2019*

Question 2: Innovation is the key to prosperity in an age of new technologies

Extract 5: United Kingdom (UK) government to bring forward ban on new petrol and diesel cars

Emission of nitrogen oxides (NO_x) from car exhausts regularly exceeds safe levels. Studies have linked it to heart and lung diseases as well as other conditions. Petrol and diesel vehicles produce the overwhelming majority of roadside NO_x gases. The removal of petrol and diesel vehicles from the roads should help bring air quality within the guidelines set out by the World Health Organisation (WHO).

The UK government is to bring forward a ban on new petrol and diesel cars to 2035, five years earlier than previously planned. About a third of NO_x emissions in the UK come from transport. The ban on conventional car sales will help move the UK towards the net-zero emissions target that the government announced in 2019. The Committee on Climate Change (CCC), which advises the UK government, said if other countries followed the UK, there was a good chance of staying below the recommended maximum global temperature rise of 1.5 degrees Celsius (°C) by 2100. This is considered the threshold to avoid dangerous climate change. However, banning new petrol and diesel cars will not be enough on its own. The government will also need to address the emissions coming from domestic and industrial energy use.

The CCC believes the price of electric cars will be similar to that of petrol or diesel cars by 2024–25. But for now, electric cars are more expensive than their petrol or diesel substitutes. This is because battery technology is still developing and production requires expensive raw materials. The UK government currently provides a £3500 subsidy for the purchase of a new electric car, but for many consumers, the price is still too high.

Some experts say a more radical approach, such as no sales tax on electric cars, would create a large increase in demand. A scheme like this exists in Norway, where there was a significant increase in the sales of electric cars in 2019.

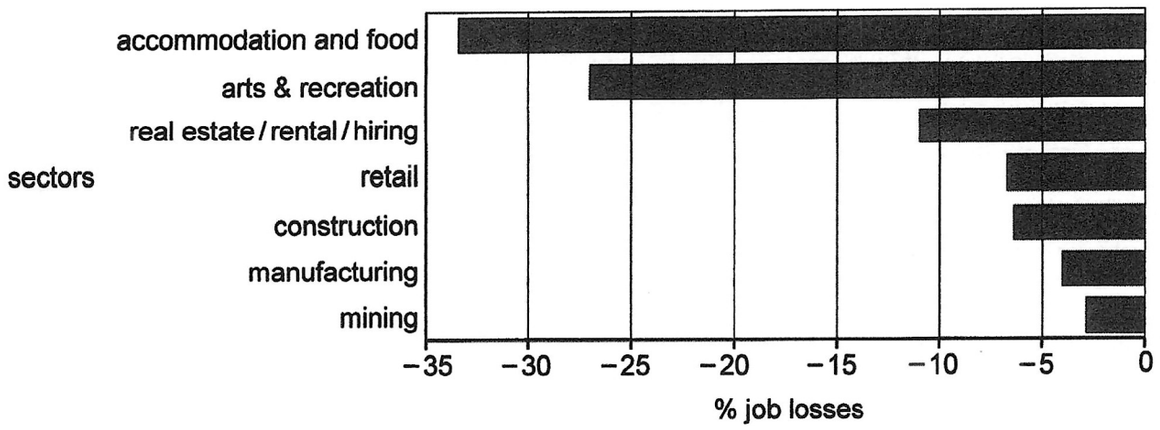
Source: BBC News, 4 February 2020

Extract 6: The German car industry faces challenges

A crisis is sweeping the German car industry. Two leading German car producers announced more than 20 000 job losses in the first signs of the human cost of the sector's transition from petrol and diesel driven cars to electric cars. Volkswagen is seeking to become a world leader in battery-powered cars. It is estimated that the German car industry, which directly employs 830 000 people and supports a further 2 million in the wider economy, will be forced to invest some 40 billion euros (€40billion) into battery-powered technologies over the next three years. German car manufacturers and component suppliers have announced that around 50 000 jobs will be lost this year, as their traditional businesses become less profitable. The global economic slowdown and the US–China trade war have forced manufacturers to revise their sales projections downwards. They had expected to sell in excess of 100 million vehicles in 2019. With just a few weeks left in the year, that figure is likely to be closer to 90 million.

Increasingly localised production by car makers in China and the US to avoid tariffs has also reduced the exporting power of German-based firms, which are left dependent on the more fragile European market. To make matters worse, demographic change could permanently reduce the number of new car buyers. The German car industry needs to adapt and change faster. In the next decade, almost a quarter of a million jobs in car production will be lost in the country. Car sales in China, which helped Germany's biggest brands survive the financial crisis, have slowed for 17 months in a row, drastically reducing a key source of revenue at the precise moment it is needed to fund new technologies.

Figure 2: Australian job losses since coronavirus restrictions, March to April 2020



Source: Australian Bureau of Statistics

Extract 4: Singapore eases monetary policy as its economy is hit

The Monetary Authority of Singapore (MAS) eased monetary policy today as the city-state, seen as a reliable indicator of the health of global trade, heads for a deep recession due to the coronavirus pandemic. The easing matches moves made by other countries and comes after data last week showed the city-state suffered its sharpest economic contraction since the global financial crisis of 2008–09.

The MAS allowed the Singapore dollar to weaken against a basket of currencies of its major trading partners. Instead of using interest rates, trade-reliant Singapore manages monetary policy by letting the Singapore dollar rise or fall.

A spokesperson for the MAS said 'Major uncertainty remains. The recovery in the global economy will depend on the course of the pandemic and the effectiveness of policy responses'.

The MAS was supposed to issue its next policy statement in April but brought it forward as the country suffers from the economic impact of the virus. Singapore is one of the world's most open economies, and is usually hit hardest and earliest during any global shock. Gross Domestic Product (GDP) shrank by 2.2% in the first quarter compared with the previous year. This was the worst decline since the 2008–09 financial crisis, according to figures released last week by the Ministry of Trade and Industry (MTI). The growth forecast has been downgraded, and GDP is projected to fall by up to 4% this year.

With global demand severely reduced by business closures, a halt in air travel and other measures to contain the virus, Singapore's easing of monetary policy is more to reflect the current economic climate than to support exports. An economist expressed the view that we cannot have a strong Singapore dollar when the global economy is in deep recession.

Source: Asia Times, *30 March 2020*

Answer **all** questions.

Question 1: Economic problems caused by drought and the coronavirus (COVID-19) pandemic

Extract 1: Australia's bushfires will significantly reduce consumer confidence and harm industries such as farming and tourism

Australia is suffering from devastating bushfires that will cause severe economic damage. The fires will further reduce Australia's already weak consumer confidence and increase the likelihood of a cut in interest rates. The fires are causing damage to the economy through increased air pollution and direct harm to industries such as farming and tourism.

In the past, bushfires had tended only to hurt local economies directly in the path of the flames, but now the risk of there being indirect macroeconomic effects is high, given the scale of the fires. The direct effect of the fires on local industries comes on top of the pain of Australia's lengthy drought that caused damage to crops and put upward pressure on prices of fresh fruit and vegetables. Tourism has also been badly affected by the bushfires in what is normally peak season. Tourism representatives say the rebuilding will cost hundreds of millions of Australian dollars (A$). Visitor numbers are significantly down as smoke haze and uncertainty about safety keep local and international travellers away.

The indirect damage to the economy will be significant. This includes air pollution, which has affected 30% of the population and will cause reduced worker productivity, increased health spending, and lower crop yields.

The devastating impact of the fires will also mean that the already fragile domestic consumer confidence will be further harmed. Australian consumers were already reducing their discretionary spending and now the widespread pollution and devastation are further deterrents.

The Australian central bank is likely to cut the official interest rate from the already record low level of 0.75%. Direct spending by the government on the economic recovery effort is likely to be more effective. Fiscal measures can target the particular areas concerned, whereas monetary policy tends to have a more general impact. The government has so far pledged A$2 billion over the next two years for measures to deal with the damage caused by the bushfires. The previous policy commitment to running a budget surplus is now seen as unrealistic.

Source: Ben Butler, The Guardian, *8 Jan 2020*

Extract 2: Coronavirus: Australia set for first recession in three decades

Official figures show that the Australian economy contracted by 0.3% in the first three months of the year, amid bushfires and the early stages of the coronavirus outbreak. Economists expect data for the second quarter to confirm that shutdowns have pushed the country into recession. It comes even after the government and central bank took measures to support the economy.

The latest Gross Domestic Product (GDP) figures highlight that the economy was struggling from a devastating bushfire season, a slowdown in tourism and weak domestic demand even before the virus-related restrictions started. In March 2020, the Reserve Bank of Australia cut the interest rate to a new record low of 0.25%. A rise in the Consumer Price Index (CPI) of 2.2% in the 12 months to March 2020 has made the real interest rate negative. The Australian government has also pledged hundreds of billions of Australian dollars to help support businesses and individuals and has signalled that more stimulus measures would be announced soon.

Source: BBC News, 3 June 2020